PF-CT Design Basis — rev L4.2

SUPERSEDED-FIGURES NOTICE (2026-11-02 cleanup). This rev L4.2 package (28 Oct 2026) is kept as the working record. Several figures in it are superseded: current masses are in data/mass_ledger.json (loco 20,404.2 kg pending freeze; tender 8,993.3 kg TB-3.15e; coaches F7 r5b 13,414.37 / 10,693.71 kg loaded), and current rulings are on status.html (sand system DELETED - sand dome is cosmetic trim only; 60 lb bolted relay rail; Loco No. 27; 4 coaches; C4 wheelchair boarding; level layout F; T04 spring brake is parking/holding only; PX-1 = ASCO 8327G041 in heated box; DF1 bogie on HOLD, baseline F7 r5b on L3 trucks; 6 mph governed operating limit, 6.5 mph HMI warning, 7 mph independent trip, 3 mph max if cameras fail; dead-loco tow ≤ 1 mph). Rows marked [SAND DELETED] are historical and must be re-run without sand. Every row is a desk calculation; NOTHING has been physically tested. Not PE-reviewed.

PROFESSIONAL ENGINEER - REVIEW / STAMP
PE stamp and signature

Name: _____
NV PE No.: _____
Date: _____
Disposition: ☐ Approved ☐ Approved as noted ☐ Revise and resubmit

1 Scope

36 in (914.4 mm) gauge amusement train for Pulsipher Farms, Logandale NV: steam-outline diesel-hydraulic locomotive (8 × Ø1,450 spoked drivers on two inside-frame powered trucks + Ø650 leading truck), tender T2 carrying the traction engine, hydraulic pump and coach generator, and 4 enclosed coaches (3 regular + 1 accessible) with emergency stairs both sides.

2 Codes and authority

• Clark County Code Ch. 22.16 (Ord. 5378): Nevada PE stamps design and calcs → third-party inspection (County-approved QA agency) → permit/operating inspection. County items are handled by the owner's representative.
• ASTM F2291-23 (design), F1193 (manufacture/quality), F770 (operations/maintenance), F3598 (as applicable); AISC 360-22 (ASD); AWS D1.1/D1.3; Poclain MS05 catalog + Application Engineering review; ANSI/ASME B29.1 roller chain; ISO 281; ISO 4413; NFPA 70 (2023) Art. 400/445/250/522/525; NFPA 101 §7 (egress reference); ASCE 7-22 (wind); IBC 2021 §1607/1011 (walking surfaces, guards, stairs - reference); 2010 ADA §234 (accessible coach).

3 Operating envelope

Item	Value	Basis
Max operating speed	6 mph GOVERNED (hard limit: H1P053 pump stroke stop sealed at 42 cc + hard tamper-sealed 2,600 rpm engine limit + PLUS+1); independent overspeed trip 7 mph → emergency brake; hardware capable of ~7.6 mph (margin only)	OWNER-SELECTED 28 Oct 2026 - PE to confirm
Design grade	2% GOVERNING design grade (start, hold and drag braking); max 2.0% on tangents, curve-compensated (equivalent grade incl. curve resistance ≤ 2.0%)	owner plan + route redline 28 Oct 2026
Min. curve radius	100 m design basis; working route likely Alt F R80 (owner pending) - E R70 / F R80 pass the L4 route recheck, D R60 fails L/V	owner plan + route L4
Wind	run ≤ 40 mph gust; parked 100 mph (ASSUMED) tied down	R2 decision
Ambient	20 °F to 110 °F; cooling equipment at 43 °C	plan CL1.1
Occupancy	32 per regular coach, 13 accessible + 1 crew per car	plan

4 Masses (rev L4.2)

Vehicle	kg	Note
Locomotive	19,934 (SUPERSEDED → 20,404.2 pending freeze, data/mass_ledger.json)	L4.2: safety-chain parts 32.2 kg (2 × PL44.5 A572-50 lugs faced to 42.5 in the jaw zone + PL16 diaphragms, 1-1/8 in G-2130 shackles, Crosby S-1325A couplers; X11150; replaces the PL40 10.2 kg), Ø120 traction pins + GE120ES-class housings +2 × 22 kg (X4075/7275), sanders on both trucks (rear boxes 46 kg body-mounted X6400 + 8 kg nozzles X7275, front upgrade 6 kg), float bypass block 14 kg, T04 orifices / 2-channel vent switch / 96-tooth pony wheel 7 kg, brake-release piping 38.5 kg GROSS (Loco Bot E7, replaces the 5.0 kg net), MR feed JL-A→BP-1 11 kg (X8190), 20 L release reservoir 14 kg; L4.1: + T04 release circuit rev (6.0 L + 1.0 L accumulators, hand pump, shuttles; steering 7) 20 kg X5675, + low traction pins 2 × 55 kg (X4075/7275, ASSUMED), Loco Bot close-out 21.3 kg (saddle gussets 12.7, SD-1 caps 3.2, SD-2 clamps 5.4), + HL-1 headlamp 25 kg (upper bound) X175, TC/valve package 28 kg X5675, ballast clamp bars 21.1 kg X4075, hub 31.4 kg/axle (Ø350 flange), Loco Bot ledger 01:57 (barrel 2,076.2, saddles 195.7, bell 14.3, whistle 4.9, slats 21.9, L-107 -1.0); no flow divider. Loco = RT1-corrected base 16,657 (INCLUDES the L3 3 mm hood 898 kg) + L4 drive delta + Loco Bot shell ledger rev 2 (Steering 17): L-108 barrel 2,280.6 incl. saddles (S4N) - 898 hood, L-109 smokebox 227.9 IN FULL (the L3 smokebox front left with the ~1,000 kg L3 cab/front - no double count), L-110 fittings 51, L-107 P1-L net +28.2 + SS± HSS3x3x1/8 +9.1, cab L-111 C1 813.9 @X9283 replacing the ~1,000 kg L3 cab/front, fit-out 6, FT-1 15.6, BT-1 battery 68.2 in the Y- front pocket (base also INCLUDES the L3 PL6 deck 1,248) + front ballast plates trimmed 1,056 -> 150.8 (2 x 75.4); pilot slats 21.9 and JL plates 9.7 (ledger rev 2) + pony own ballast 1,250; jumpers 40 kg in train mass only; L4 drive delta 571 kg (Poclain/Tsubaki/Martin catalog masses; 45T split ring + web ~50 kg ASSUMED, hub 31.4 kg CALC (Ø350 flange), case, disc ASSUMED; NO flow divider) [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
Tender T2	8,706.5 (SUPERSEDED → 8,993.3 TB-3.15e)	tender_mass_T2.json
Coach regular loaded / empty	13,200 / 8,040 (SUPERSEDED → 13,414.37 / 10,444.37 F7 r5b)	ledger
Coach accessible loaded / empty	11,000 / 7,000 (SUPERSEDED → 10,693.71 / 9,069.71 F7 r5b)	empty ASSUMED
Train loaded	81,513 (SUPERSEDED → 80,334.3 sum of ledger, no crew; see flags)	W = 799.4 kN

5 Load factors and allowables

• Service ×1.3 vs ASD 0.6Fy (bending/tension), 0.4Fy (shear); exceptional ×2.0 vs Fy. Drawgear 70 kN service / 200 kN ultimate. Equipment restraint 3 g long. / 2 g vert. / 1 g lat. Seats 3 g. Floors 100 psf; guards 50 plf / 200 lb; treads 300 lb. All ASSUMED where no code value - PE to confirm.
• Drive: chain pull at the 430 bar strength basis vs the maker allowable load (Tsubaki RS120-2 51.7 kN); motor radial vs Poclain dynamic chart; shafts/structure at the 430 bar strength basis (HPRV 380 + spike) with no efficiency credit. Axle seat fatigue Goodman n ≥ 1.5 with NO hardening credit, fretting Kf 2.0, heaviest axle (ASSUMED). Bearings: L10 ≥ 20,000 h; static s0 ≥ 2.
• Brakes: tare-based 12% braking ratio; emergency spring-applied fail-safe; stop from the governed 6 mph on 2% down ≤ 20 m and from the 7 mph trip ≤ 30 m (ASSUMED criteria).

6a Rev L4.2 changes (these SUPERSEDE any L4.1 text below where they differ)

• Sheet numbering (L4.2, Loco Bot shop-sheet master L-1xx): design sheet L-105 (ballast pockets, cab steps) is now L-106; design sheet L-106 (chain drive unit) is now L-103.2 (sheet 2 of L-103, final drive and motor mounting). Loco Bot shop sheets L-103, L-105 (hydraulic routing), L-106 (front ballast), L-107L, L-108, L-109, L-110, L-111, L-114, S-504, S-507 and A-101 are part of the set (G-001 index). Older calc text that cites L-105 for ballast or L-106 for the drive unit reads as L-106 / L-103.2.
• Grade basis: the TRACK is designed LEVEL (Layout F dead level 0.0 %, Rail Bot §5b FINAL); the VEHICLES are checked at 2 % as the conservative case.
• Descents: FLOAT mode - brake handle out of RUN → loop bypass valve opens and the swash ramps to zero over ≥ 3 s; NO hydrostatic retarding. Friction brakes sized for the full 2 % descent with the loaded train: 15.99 kN descent force vs 39.35 kN service friction (util 0.41); continuous-descent limit 2398 m; cooler in float 3.3 of 14 kW.

• T04: Ø1.2 orifice in each T04 port; hard lines or EN 856 4SH hose downstream of the choke; 2-channel vent switch. Release fed from the H1 CHARGE circuit (M3 port) → check + 10 µm filter → CDK 20 bar reducing/relieving → normally-vented 3/2 → Ø2.4 choke + 1.0 L damping accumulator; LOOP SHUTTLE DELETED. Release 6.2 s vs the separate 8 s T04 target (met); tow / engine-off release from the 6.0 L accumulator + hand pump (5.45 L needed for 2 releases).
• Run-in (Stress Bot s4, tender TB-3.13d): worst design case 25.3 kN ≤ 33.2 kN; double failures (information only) 29.7 / 34.3 kN.
• Speed: HARD tamper-sealed 2,600 rpm engine limit + H1P053 stroke stop sealed at 42 cc: 5.96 mph loaded, 6.47 mph no-load high-idle check (< 7 mph trip); 6 mph governed; rechecked at η 0.99. MS05 shift valves FAIL-TO-1st; 2nd only through the PNOZ s30 ≥ 1.5 mph contact.
• Pressure: PL 350 / HPRV 380 bar. Loop components rated ≥ 420 bar WP ≥ HPRV 380; the transient spike (~430 bar) is covered by the ISO 18752 / ISO 6605 impulse rating (133 % of WP); STRENGTH cases are checked at 430 bar. The Danfoss overshoot figure is information only, not gating. Loop hose Manuli DIAMONDSPIR H10018025 (ISO 18752-C, 560 bar WP, OD 41.2, min bend 350 mm ≤ the tender's tightest 394.3 mm at R70, 2.91 kg/m). Tow valve across A-B: HYDAC KHB3K DN06 class, 500 bar, sized for the 2 mph tow flow (DN25 310 bar is NOT allowed on the loop).
• 15T sprocket: MADE TO ORDER in 4140 Q&T 28-32 HRC, DIN 5480 N55x3x17, ≥ 58 mm engaged (stock C1045 hub 1.03 NG at 430 bar × KA 1.5).
• Chain: pull at 430 bar 35.2 kN; breaking-strength certificate ≥ 326 kN per lot (same factor as the old 302.5 kN at 400 bar). Standard Tsubaki RS120-2 296 kN passes at HPRV 380 (287.4 kN) but NOT at the 430 bar strength basis - higher-strength 120-2 = vendor batch, or PE acceptance of SF 8.41 on ultimate.
• Flange (FINAL, train-wide): withdraw at 28 mm, re-profile to ≤ 27 mm; joint rail-head wear limit 2 mm (rail height condemn ≤ 86.9 mm at the bolts; 6 mm only > 1 m from rail ends); joint bolt protrusion ≤ 10.0 mm measured at install (flat bars ≤ 12.0 mm, 3/4 in B18.10 oval-neck bolts H ≤ 12.7 mm); pony 22 mm.
• Traction pin Ø120 4140 Q&T + GE120ES-class bearing (+22 kg per pin); centre-plate friction (laminated elastomer pad + PTFE yaw slide, µ ≤ 0.08, ASSUMED) included in the start margin.
• Sanders on BOTH powered trucks with level switch, flow proof and hill-hold. Forward 2 % start margin 1.112 (≥ 1.10). Reverse: margin 0.899 at 2 %; REVERSE FULL-TRAIN START LIMIT 1.20 % (governed by the 28.9 kN reverse pull cap of record with 1.10; adhesion alone gives 1.35 %). [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
• K2-a of record: 25.89 kN (forward PLUS+1 TE command limit 33.0 kN = 1.01 × need) and 25.74 kN at the 28.9 kN reverse cap, vs 26 kN; fixed with NB-1 nose ballast 59.8 kg (PL25.4 600 × 500) in the smokebox floor on the S1 saddle at X450 (Loco Bot to detail on L-109).
• Pony reference: ~96-tooth target wheel, 2 Hall sensors in quadrature, pony/axle agreement check; PNOZ s30 kept independent.
• Release targets: T04 ≤ 8 s (separate, met); air release ≤ 4 s (2.36 s worst branch, MR ≥ 5.6 bar; one release takes the ~190 L coupled MR 6.0 → 5.68 bar; 20 L local reservoir deleted); coach release indication DUAL-CHANNEL per Electro E3.6h (trainline J-CC 22/23 + 34/35, Pilz B0 test pulses, PL d Cat 3, daily force-guided relay test). PE to confirm the fault exclusion for the single pressure element per switch and the test-pulse loop length.
• Brake air (Loco Bot E7/E8): trunk and MR feed 3/4 in 316 25 × 2.5, MR feed 8.75 m; SAE J1402 hoses marked ≥ 225 psi WP / ≥ 900 psi burst; QEV option A (delete) REJECTED - spring-apply 2.13 s > 2.0 s although its 7 mph stop passes (21.7 m); option B KEPT: maker-certified ≥ 15 bar 1/2 in QEV (vendor batch), apply 0.94 s, 7 mph / 2 % stop 20.1 m vs 30 m; piping 43.7 kg gross.
• Safety chains: lugs PL44.5 A572-50 faced to 42.5 mm with 1-1/8 in G-2130 shackles; each chain = 2 × G-2130 + 2 × Crosby S-1325A (fallback A-1342N) + 1-2 links 5/8 G100; coupler WLL ≥ chain WLL (vendor batch check, Loco Bot E8 PASS). BP trunk exception to L-114 note 3: ≥ 25 mm to the chain cases and ≥ 75 mm from hot lines.
• Wind and stop: wind at R80 is covered by the interim rule (25 mph wind limit on R ≤ 80; no full-pull starts with a lubricator out); the 10 mph stop is information only.
• Ballast clamp P-106-1 (4 × FB100x16 × 475, 23.9 kg, inside the pocket on M16 set screws) and P-106-2 (60 mm pocket walls) APPROVED; P-103-1 motor hanger L-weld APPROVED, fatigue check pending at Loco Bot.
• Electro E3.6f: coach and tender jumpers are supported loops (saddles at Z720, worst low point Z286 > 150 at R70 buff); loco-tender cords bottom at ≥ Z1060; trainline tap QO2225SL lugs in the QO142.
• Tender TB-3.13d 8,706.5 kg, CG [2422.7, 11.3, 1110.4] (incl. the tender-end shackles; +7.9 kg if the A-1342N fallback coupler is used); full brake 0.671 m/s².

6 Adopted design (rev L4.2)

• Drive (rev L4.1): per powered axle one Poclain MS05 468/234 cc two-speed HighFlow motor with T04 multidisc brake and speed sensor, FRAME-MOUNTED on a slotted bracket on the transom, driving through an ANSI 120-2 duplex chain (Tsubaki RS120-2 or equal, 54 links) from a Martin D120B15 15T (OD 202; hub broached DIN 5480 N55x3x17 female on the motor shaft code 5, DIN 5480 W55x3x17 male; full spline engaged ≥ 58 mm, ~60 expected - flank 95 MPa at HPRV × 1.5, util 0.95, D2-h) to a 45T SPLIT rim ring with a bolting web (3:1), 12 × M16 cl.8.8 fitted bolts DIN 609/7968 (219 N·m dry, Nord-Lock) to the Ø350 flange of a 4140 hub (31.4 kg) PRESSED on a Ø150 raised axle seat (no keyway); ring replaceable in place without pressing , all in a 3 mm oil-bath case with a floating seal plate at the axle. Everything inside the frame between the wheels; nothing outside the wheels; no belt; no idler (slotted mount + jack screws). Pump Danfoss H1P053 (shaft G1 14T 12/24 on the SAE B flange - the engine pump-drive coupling hub is 14T 12/24, D1-u; L35: limiter 350 / HPRV 380, L4.2) with CCO control cut-off wired to the Pilz PNOZ s30 7 mph trip and an external HYDAC DF 60 charge pressure filter, 35 kW anti-stall (34.5 kW with the compressor clutched); stroke stop 42 cc + sealed 2,600 rpm (L4.2) = 6 mph governed; NO flow divider - electronic traction control in PLUS+1 (MS05 sensor per motor + 3rd pickup on the unbraked pony = ground-speed reference; slip > 10 % / 0.5 mph / 0.5 m/s² → auto-sand, destroke, ramp back; start in 1st; hill-start hold; rollback alarm); independent 7 mph overspeed relay. Loco valves (steering 7): T04 release from the loco's own loop - LP shuttle on loop A/B + 10 µm filter + 3-way reducing/relieving valve 20 bar + normally-vented 3/2 solenoid + choke Ø2.4 with 1.0 L damping accumulator + brake-line shuttle; loss of loop/charge pressure or 24 V APPLIES the T04 (after the BP vents); manual release from a 6.0 L accumulator + hand pump via a guarded valve; no 4th dry-break; tow = lockable ≥ 420 bar ball valve (L4.2) across

loop A/B with a limit switch + Poclain freewheel, max 2 mph. Rejected: L3 Dodge reducer + H1B110 + belt (fallback, calcs/L3), L2 custom case. [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
• Starting force 37.67 kN at 350 bar (η 0.90; still passes at η 0.80), adhesion-capped at 37.11 kN dry; 32.61 kN needed on 2% (~303 bar). Equal-torque cap (L4.1): dry/sanded 2% margin 1.10 (lightest axle F2 with 2 × 75.4 kg front plates + BT-1 + clamp bars); wet unsanded 0.66 → sand mandatory on wet grades/curves (auto-sand). 2% hold 2.69 mph; level 6 mph governed. [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
• Brake basis: brake pipe 5.0 bar (72.5 psi), UIC / Knorr KE distributor; WABCO IR-2 inverting relay 971 899 152 0 between the KE output and the loco spring-brake (release) chambers; ASCO 8210G035 24 VDC de-energize-to-vent valve as the brake-pipe power-loss vent. Fail-safe: loss of BP, MR or 24 V, or the 7 mph trip, applies the springs; the same s30 output drops the H1P053 CCO (pump to neutral). B-601.
• Drawgear: 5 joints; DB-800 loco-tender (kept as drawn; R60 swing 19.3°/11.0° vs 32.4°/19.2°), DB-1473 elsewhere; pins Ø60 4140 × 150 under head, 12 incl. 2 spares; 2 × 5/8 in Grade 100 safety chains per joint with Crosby G-2130 1 in shackles.
• Brakes: air, BP/MR train lines; spring-applied air-released calipers on every axle EXCEPT the unbraked pony (lead) axle (loco: 4 driven axles, Ø340 axle disc on the free side, 2 calipers per disc; tender and coach axles all braked); pony wheels Ø650 new, worn-wheel limit Ø630 - for PE sign-off, emergency on BP exhaust and on the independent 7 mph overspeed relay; parking = spring brakes + chocks; MS05 T04 brakes hold the loco alone; tow release by wind-off calipers + T04 release screws + the lockable loco tow valve (A/B) + MS05 freewheel per Poclain, max 2 mph (H1P053 HPRV bypass only for short moves).
• Loco interfaces (L-114 rev E3): DBK-1 hydraulic dry-break + air zone on the loco LEFT, X10900-11300, Y-220 to -600, Z700-1100 (no hydraulics at Y > -200 or above Z1100); electric jumpers on the RIGHT (JL-E Y+500 to +1100, Z1700-2200); air glad hands on the REAR SILL per JL-A: BP Y-650, WH (whistle, new) Y-800, MR Y-950, all at Z1950; DBK-1 dry-breaks Y-220 to -600, Z700-1100; AW-1 whistle line 1/2 in. HL-1 headlamp X175 Y0 Z3495 on 109-SBX-7 rev B.
• Front ballast: 2 × 1 in A36 plates 800 × 473 (75.4 kg each, one per pocket) + BT-1 battery 68.2 kg in the Y- pocket, held by 2 × FB100×12 clamp bars per pocket on M16 8.8 to pocket lugs (21.1 kg total; now L-106).
• Tender rev TB-3.9 8,616.3 kg, CG [2421.7, 10.6, 1110.8] (Tender Bot; was TB-3.8 8,568.8, TB-3.4 8,465.3): genset reservation X1060-2768 (≤ 1,775) × 800 × 1,000, basis PTI-38; right-wall outward-facing NEMA 3R electrical cabinet X2820-3780 (listed 200 A UL 1008 ATS with integral main; separate QOM22225NRB main, handle ≤ Z1639); custom 150 L Al fuel tank X3900-4500 behind a steel firewall at X3800; 3 × Haldex 19820 main reservoirs X3600-4440; hydraulic cooler ≥ 14 kW left wall X3455-4145. Under-deck shore transformer cradle DROPPED 50 mm (25 mm at the isolator stop under the X2500 crossmember, 100 mm to K-401) - K2-xf CLOSED. Tender brake TB-3.9: 5.825 kN full (0.676 m/s²; mixed Coremo D 2N spring sets, 25.4 mm discs, pad wear limit 5.5 mm), 4.08 kN service (0.70 × full ASSUMED), emergency limited by the relay. Max wheel (E-cg) ~11.2 kN.
• Power: 35 kW continuous 120/240 V 1-ph genset in tender + RenewAire HE05IN ERV per coach (roof pod, 62.1 OA 279/126 cfm) + staging controller.
• Truck traction (L4.1): LOW TRACTION PIN per powered truck - Ø100 4140 pin from the body centre plate down through the transom to a GE100ES spherical bearing in a PL16 housing at Z650 (h 0.65 m); the centre plate becomes a flat lined vertical bearing (55 kg/truck ASSUMED). Forward 2 % dry/sanded equal-torque margin 1.13 with weight transfer; PLUS+1 reverse pull cap 28.9 kN and full-train reverse starts only on grades ≤ 1.20 % (L4.2) (covers Ø1430 worn drivers). K2-a 25.7 kN on the RT1 axle set with transfer. [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
• Trip behaviour (L4.1): any trip (CCO overspeed, 24 V loss, s30) ramps the pump to neutral over ≥ 3 s (EDC orifices + PLUS+1); the MS05 shift valve de-energizes to 2nd; a pressure switch on the T04 release line vents the BP first and a choke in the T04 release drain delays the T04 ≥ 1.0 s and builds it over ≥ 2.0 s. Worst loco-tender run-in 23.7 kN vs 33.2 kN (R80). Direction change only at zero speed with a ≥ 1 s ramp.
• Chain (L4.1): ±22 mm hard bump stops, axlebox side float ≤ ±1 mm, seal plate ±25 / ±1 (OI-L4-5 closed); corrected slack geometry (dL = 2·span/C·dC) - sag 17-20 mm at ride height lasts to 30.8 mm bump; re-shim for wheel wear at the centre plate/side bearers only (primary ≤ 3 mm, then motor-slide reset); 2 % elongation hard limit; oil level 20-25 mm; B29.1 breaking-strength certificate ≥ 302.5 kN per lot.
• Hub (L4.1): 4140 hub SHRINK FIT ONLY, 0.15-0.20 mm diametral, heated 200-230 °C; oil pressing deleted; oil-injection groove kept for removal only (SF 1.87 at HPRV with a 30 K hot hub).
• Speed (L4.1, SUPERSEDED by §6a - 42 cc + sealed 2,600 rpm): H1P053 stroke stop sealed at 38 cc: 5.91 mph at the 2,850 rpm high idle (ASSUMED), ~5.4 mph at rated 2,600; one s30 overspeed channel on the UNBRAKED pony tone ring.
• Operating rules (L4.1, descents SUPERSEDED by §6a FLOAT mode): friction brakes control descents; sand mandatory for wet 2 % starts; high-rail gauge-face lubricators on ALL R ≤ 80 (top of rail dry); dead-loco tow: tow valve open, T04 released, motors to 2nd, max 2 mph. [SAND DELETED 2026-11 - historical row, re-run without sand; see status.html]
• Pony (L4.1): spring rate ≤ 1.23 kN/mm per wheel, static deflection ≥ 5 mm.
• Loco body (L4.1, Loco Bot items): saddle web gussets 8 × PL10 45×150 per saddle at Y±535/±685; B1 HB16 holes X480/600 (28 mm from the sill weld toe), 28 HB16 total; SD-1 cap re-cut to Y±50 for the Y±90 HB16; SD-2 hold-down by 4 PL16 clamp plates under the end-sill flange (no sill holes); ballast 1 plate per side + 4 × FB100×12×560 clamp bars (rod nuts 30 N·m); bell yoke Ø32 A4 with Ø20 journals (F-13 through pin deleted); SS± tabs M12×100 8.8 through-bolts with crush tubes; rivnut production pull test 6.0 kN.
• Loco ledger (L4.1 steering 9): + safety-chain lugs 2 × PL40 A572-50 (CJP to the rear headstock, ~3.2 kg each) + 2 × PL16 internal diaphragms (~1.9 kg each ASSUMED) = 10.2 kg at X11150; + 5.0 kg brake release-air upgrade (ASSUMED). Lug CJP welds UT 100 % (S-503).
• T04 release (L4.1 steering 7, SUPERSEDED by §6a - charge feed, shuttle DELETED): NO 4th dry-break. [was: LP-select shuttle on loop A/B] -> 10 µm filter -> 3-way reducing/relieving valve 20 bar (fully open below setting, so the brake line follows a falling charge pressure) -> normally-vented 3/2 solenoid -> choke Ø2.4 + 1.0 L damping

accumulator (apply delay 1.01 s, rise 2.97 s; release ~4.8 s) -> brake-line shuttle -> 4 T04. Manual release: check -> 6.0 L bladder accumulator (pre-charge 13.5 bar, 2 releases) + 12 cm³ hand pump -> guarded padlocked valve with limit switch (traction inhibit). Charge switch 17 bar falling vents the BP first. Loss of loop/charge or 24 V applies the T04. T04 release volume and pressures ASSUMED (0.10 L/motor, full release 15 bar, max 30) until Poclain data.

• Wheels (L4.1 steering 7): driver condemning diameter Ø1430 (new Ø1450) ADOPTED - top speed 5.83 mph at high idle (governor unchanged), trip set in axle rpm for Ø1450 (6.90 mph at Ø1430), TE 38.2 kN at 350 bar, forward 2 % margin 1.116, reverse cap 28.9 kN -> rule ≤ 1.20 % (L4.2), lower outline 245 vs 150, loco brake 0.70 m/s² (run-in +0.2 kN). FLANGE-HEIGHT limit 28 mm / ≤ 27 (L4.2 FINAL) on drivers and pony (35.0 mm at Ø630/Ø1430 worn is NOT permitted; ASCE 40 joint-bolt heads ~32.5 mm below TOR); frog/guard flangeway depth ≥ 36 mm to confirm on the owner track drawings.

• Brake air (L4.1 steering 8/9): loco calipers 8 × 1.0 L = 8.0 L, 48 L free air and 0.98 bar MR drop per release, recharge 17.8 s at 14 cfm. LOCO RELEASE ≤ 4.0 s WORST CASE (command → every caliper ≥ 5.1 bar; supply drooped to 5.6 bar, isothermal, 1.0 s signal/relay allowance ASSUMED): release path total sonic conductance C ≥ 3.0 dm³/(s·bar) (required 2.5) gives 3.46 s; parts budget C 5.26 gives 2.4 s. Parts: relieving release regulator 1/2 in C ≥ 10 set 6.0 bar; 1/2 in brake relay valve C ≥ 8 piloted by the WABCO 971 899 152 0 inverting relay (fail-safe logic unchanged); 3/4 in trunk ID ≥ 19 ≤ 12 m equivalent C ≥ 8 (full-bore valves only); per axle 1/2 in hose ID 12.7 ≤ 2 m C ≥ 2.0 to a tee at both calipers; quick-exhaust per axle; release-proof switch 5.1 bar → traction enable. Loco Bot line lengths and fittings to confirm trunk/branch C. Train: 14.0 L calipers, 84 L free air per release, 3.55 cfm steady vs 14 cfm (25 %).

• Cross-references (L4.1): tender TB-3.9 CLOSED (8,616.3 kg, brake 0.676 m/s² vs loco 0.690, cradle -50 mm, K2-xf closed; caliper air train 14.0 L, 84 L free air per release, 3.55 cfm steady vs 14 cfm; ERV roof 3,450 static, kinematic checked vs K-401 4,300); genset/LED load CLOSED (Electro E3.4 already carries the 1.26 kW LEDs in 30.99 kW running / 32.49 kW peak hot; decorations shed first above 33 kW); lights per Electro (HL-1 25 kg carried).

7 Deliverables

• PF-CT_PE_calc_package.pdf (numbered calc index, 12 subsystems); PF-CT_R2_drawing_set.pdf (19 sheets, index on G-001); this basis; Assumptions & Open Items; Reviewer cover memo; CAD STEP/GLB per assembly (cad/).